

Experiment 3

Capturing and Analyzing Network Packets

Lab 3 Objectives:

- Understand the structure and functioning of the OSI and TCP/IP models.

Prerequisites: The prerequisites of this lab are-

- Basic understanding of computer networks and communication protocols.
- Familiarity with OSI and TCP/IP models.
- Experience with network monitoring tools such as Wireshark.

Outcomes: By the end of this lab, students will be able to:

- Capture and interpret network packets using Wireshark.
- Analyze how different protocols map to OSI and TCP/IP layers.
- Understand the flow of network traffic and troubleshoot network-related issues.

Description: In this experiment, students will use Wireshark, a network protocol analyzer, to capture real-time network packets and analyze their contents to understand how data flows through the OSI and TCP/IP models.

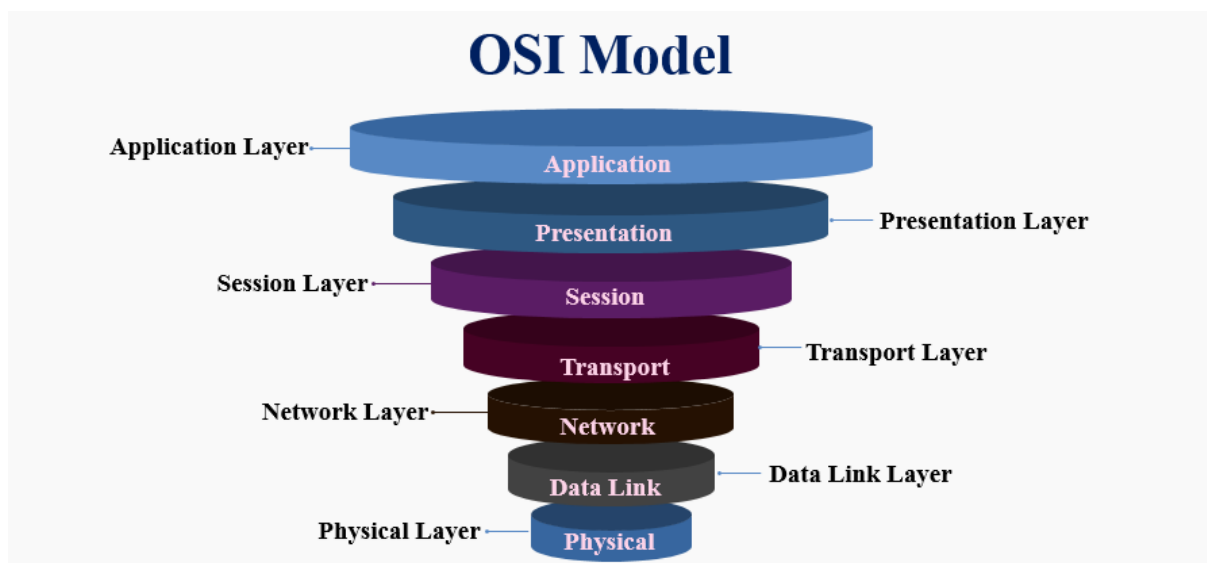
Part 1: Understanding OSI and TCP/IP Layers

OSI Model

The **OSI (Open Systems Interconnection)** model is a conceptual framework used to understand and design networks. It divides network communication into **7 layers**, each with specific functions:

1. **Physical Layer:** Deals with the physical connection between devices, including cables and signals.
2. **Data Link Layer:** Handles error detection and frames data for transmission between adjacent nodes.
3. **Network Layer:** Manages logical addressing and routing of data packets.

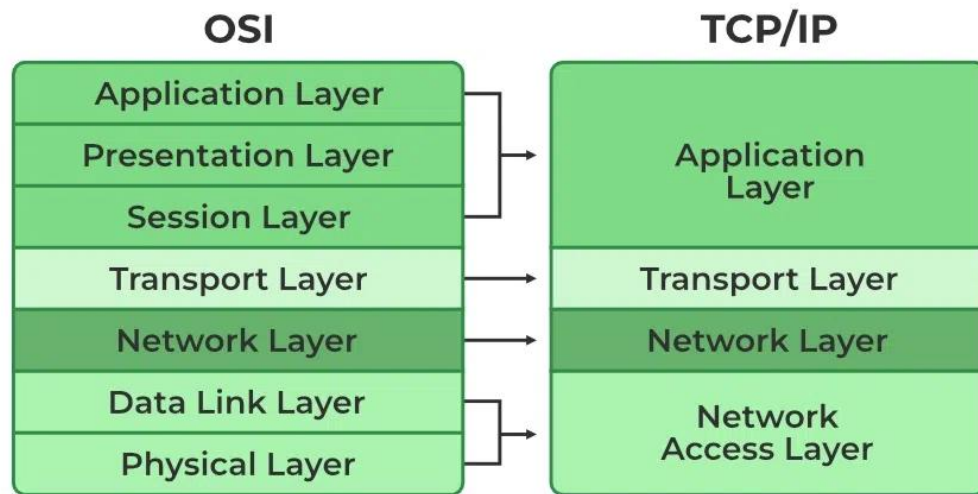
4. **Transport Layer:** Ensures reliable data transfer with flow control and error recovery.
5. **Session Layer:** Manages sessions or connections between applications.
6. **Presentation Layer:** Translates data formats and handles encryption/decryption.
7. **Application Layer:** Interfaces directly with user applications and provides network services (e.g., HTTP, FTP).



TCP/IP Model

The **TCP/IP model** is a practical, simplified model used in real-world networks and the internet. It has **4 layers**:

1. **Network Interface Layer:** Combines OSI's Physical and Data Link layers, managing hardware addressing and transmission.
2. **Internet Layer:** Handles logical addressing and routing using the IP protocol.
3. **Transport Layer:** Provides end-to-end communication with protocols like TCP (reliable) and UDP (unreliable).
4. **Application Layer:** Supports network services and applications such as HTTP, FTP, SMTP, and DNS.



Protocols Analyzed in Packet Capture

When capturing network traffic using tools like Wireshark, the following protocols are commonly analyzed to understand network communication:

- **HTTP (Hypertext Transfer Protocol):** Used for web browsing (Application Layer).
- **TCP (Transmission Control Protocol):** Ensures reliable, ordered data transfer (Transport Layer).
- **UDP (User Datagram Protocol):** Provides faster but connectionless data transfer (Transport Layer).
- **IP (Internet Protocol):** Routes data packets between devices (Internet Layer).
- **ARP (Address Resolution Protocol):** Resolves IP addresses to MAC addresses (Network/Data Link Layer).
- **ICMP (Internet Control Message Protocol):** Used for diagnostic tools like ping (Internet Layer).

Capturing Network Packets Using Wireshark

When Wireshark is opened, it displays a list of active network interfaces. In my case, my laptop was connected via Wi-Fi, so I selected the Wi-Fi interface.

Image

The active network interfaces are shown in Wireshark, including my Wi-Fi connection.

Analyzing Captured Packets in Wireshark

Filtering Packets

Wireshark provides a filter bar at the top, allowing users to filter packets by specific protocols. Common filters include:

- `http` — Displays HTTP traffic
- `tcp` — Shows TCP segments
- `udp` — Shows UDP packets
- `icmp` — Displays ping requests and replies

By typing the protocol name and pressing Enter, only packets related to that protocol are displayed.

Examining Packet Details

Clicking on any packet reveals detailed information below, including the source and destination IP addresses, port numbers, and protocol headers such as TCP, IP, and HTTP. This helps understand how data traverses through the OSI model layers: Application → Transport → Network.

TCP Three-Way Handshake

The TCP connection is established through a three-step process known as the three-way handshake:

- **SYN:** Client sends a connection request
- **SYN-ACK:** Server acknowledges the request
- **ACK:** Client confirms the connection

Filters to identify these packets in Wireshark are:

- `tcp.flags.syn == 1 and tcp.flags.ack == 0` — SYN packets
- `tcp.flags.syn == 1 and tcp.flags.ack == 1` — SYN-ACK packets
- `tcp.flags.ack == 1 and tcp.flags.syn == 0` — ACK packets

Wireshark screenshot displaying the TCP three-way handshake with SYN, SYN-ACK, and ACK packets.

Time	Source	Destination	Protocol	Length No.	Info
0.000000	192.168.226.42	224.77.77.77	UDP	155	1 52300 → 12177 Len=113
1.977096	142.250.67.202	192.168.226.42	UDP	74	2 443 → 63848 Len=32
1.977538	192.168.226.42	142.250.67.202	UDP	76	3 63848 → 443 Len=34
2.016138	192.168.226.42	224.77.77.77	UDP	155	4 52300 → 12177 Len=113
4.024897	192.168.226.42	224.77.77.77	UDP	155	5 52300 → 12177 Len=113
5.497360	192.168.226.42	104.18.32.47	TCP	55	6 53777 → 443 [ACK] Seq=1 Ack=1 Win=254 Len=1 [TCP PDU reassembled in 32]
5.572218	104.18.32.47	192.168.226.42	TCP	66	7 443 → 53777 [ACK] Seq=1 Ack=2 Win=19 Len=0 SLE=1 SRE=2
6.028337	192.168.226.42	224.77.77.77	UDP	155	8 52300 → 12177 Len=113
7.051497	192.168.226.42	64.233.170.188	TCP	55	9 53533 → 5228 [ACK] Seq=1 Ack=1 Win=253 Len=1
7.179959	64.233.170.188	192.168.226.42	TCP	66	10 5228 → 53533 [ACK] Seq=1 Ack=2 Win=1048 Len=0 SLE=1 SRE=2
8.043137	192.168.226.42	224.77.77.77	UDP	155	11 52300 → 12177 Len=113
8.922103	192.168.226.42	150.171.27.11	TCP	55	12 53767 → 443 [ACK] Seq=1 Ack=1 Win=255 Len=1
9.061624	150.171.27.11	192.168.226.42	TCP	66	13 443 → 53767 [ACK] Seq=1 Ack=2 Win=16381 Len=0 SLE=1 SRE=2
10.056273	192.168.226.42	224.77.77.77	UDP	155	14 52300 → 12177 Len=113
11.656950	192.168.226.42	51.104.15.253	TCP	55	15 53770 → 443 [ACK] Seq=1 Ack=1 Win=254 Len=1
11.919755	51.104.15.253	192.168.226.42	TCP	66	16 443 → 53770 [ACK] Seq=1 Ack=2 Win=16381 Len=0 SLE=1 SRE=2
12.015247	142.250.67.202	192.168.226.42	UDP	74	17 443 → 63848 Len=32
12.015630	192.168.226.42	142.250.67.202	UDP	76	18 63848 → 443 Len=34
12.063655	192.168.226.42	224.77.77.77	UDP	155	19 52300 → 12177 Len=113
13.054546	192.168.226.42	3.233.158.24	TCP	55	20 53801 → 443 [ACK] Seq=1 Ack=1 Win=252 Len=1
Frame 7: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\NPF_{7EC3D0958-37} Ethernet II, Src: ea:6b:3a:d5:9e:b8 (ea:6b:3a:d5:9e:b8), Dst: Intel_a0:10:13 (dc:45:46:a0:10:13)					
Internet Protocol Version 4, Src: 104.18.32.47, Dst: 192.168.226.42					
Transmission Control Protocol, Src Port: 443, Dst Port: 53777, Seq: 1, Ack: 2, Len: 0					

Hex dump view:

```

0000 dc 45 46 a0 10 13 ea 6b 3a d5 9e b8 08 00 45 00 .EF...k:...E.
0010 00 34 ca b3 40 00 35 06 4f fc 68 12 20 2f c0 a8 .4 @ $ O h / .
0020 e2 2a 01 bb d2 11 29 70 a8 bd 6d 69 25 d5 80 10 *....p...mi%..
0030 00 13 ee e0 00 00 01 01 05 0a 6d 69 25 d4 6d 69 %.....mi%mi
0040 25 d5
    
```